

MODULE 4.4 PRACTICAL PROJECT ASSIGNMENTS

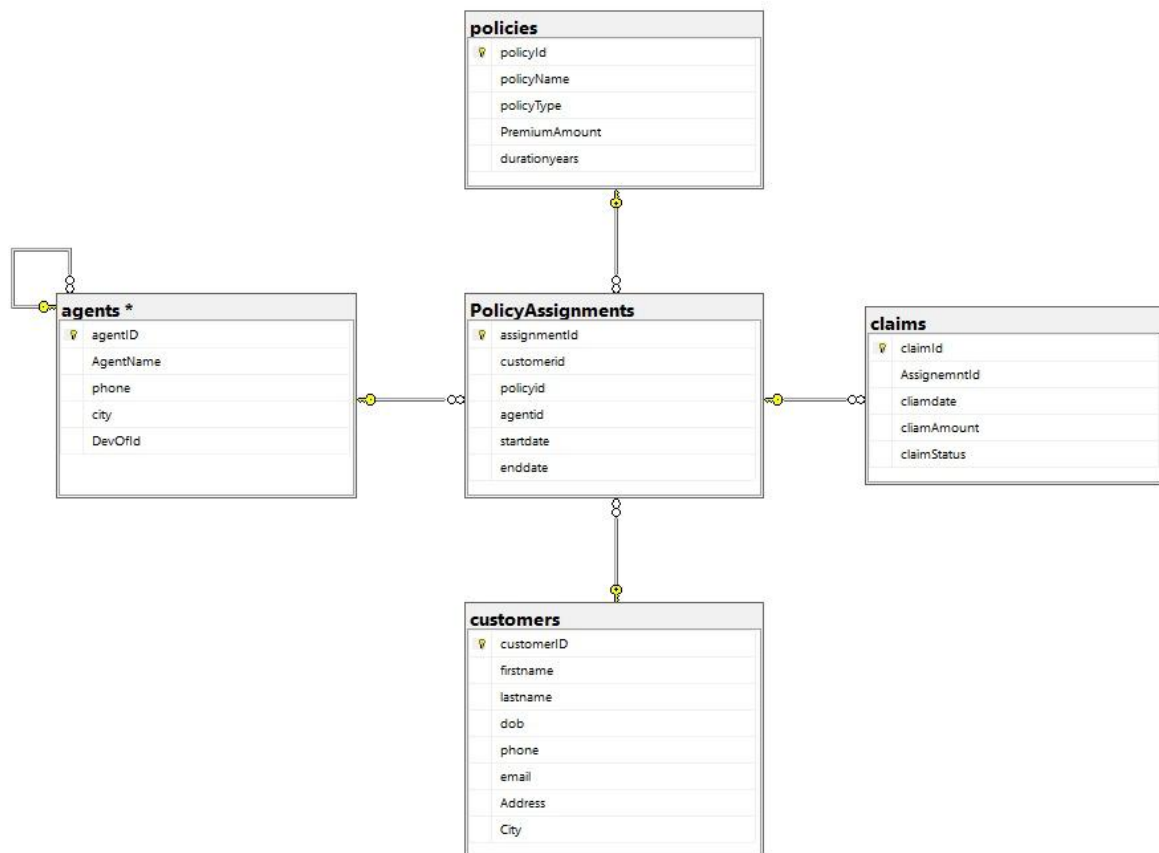


Fig 4.4.1 : InsuranceDB Schema

4.4.1 Create Database

create database InsuranceDB;

use InsuranceDB;

4.4.2 Create table commands for all the tables with constraints, relationships etc.

Customers Table

CREATE TABLE Customers (

CustomerID INT,

FirstName VARCHAR(20),

LastName VARCHAR(25),

DateOfBirth DATE,

Phone VARCHAR(10),

Email VARCHAR(20),

PRIMARY KEY (CustomerID)

);

Policies Table

CREATE TABLE Policies (

PolicyID INT,

PolicyName VARCHAR(20),

PolicyType VARCHAR(30),

PremiumAmount MONEY,

DurationYears INT,

PRIMARY KEY (PolicyID)

);

Agents Table

CREATE TABLE Agents (

AgentID INT,

AgentName VARCHAR(20),

Phone VARCHAR(10),

City VARCHAR(15),

PRIMARY KEY (AgentID)

);

PolicyAssignments Table

CREATE TABLE PolicyAssignments (

AssignmentID INT,

CustomerID INT,

PolicyID INT,

AgentID INT,

StartDate DATE,

EndDate DATE,

PRIMARY KEY (AssignmentID),

FOREIGN KEY (PolicyID) REFERENCES Policies(PolicyID),

FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),

FOREIGN KEY (AgentID) REFERENCES Agents(AgentID)

);

Claims Table

```
CREATE TABLE Claims (  
    ClaimID INT,  
    AssignmentID INT,  
    ClaimDate DATE,  
    ClaimAmount VARCHAR(10),  
    ClaimStatus VARCHAR(10),  
    PRIMARY KEY (ClaimID),  
    FOREIGN KEY (AssignmentID) REFERENCES PolicyAssignments(AssignmentID)  
);
```

Payments Table

```
CREATE TABLE Payments (  
    PaymentID INT,  
    PolicyID INT,  
    PaymentDate DATE,  
    Amount MONEY,  
    PRIMARY KEY (PaymentID),  
    FOREIGN KEY (PolicyID) REFERENCES Policies(PolicyID)  
);
```

Alter column type

```
ALTER TABLE Claims  
ALTER COLUMN ClaimAmount MONEY;
```

4.4.3 Insert commands for all tables.

```
INSERT INTO Customers  
VALUES  
(1, 'John', 'Darwin', '1985-12-02', '9999999999', 'johnd@gmail.com'),  
(2, 'Anita', 'Sharma', '1990-06-15', '8888888888', 'anita@gmail.com'),  
(3, 'Rahul', 'Verma', '1988-03-22', '7777777777', 'rahul@gmail.com'),  
(4, 'Sneha', 'Patel', '1995-11-10', '6666666666', 'sneha@gmail.com'),
```

(5, 'Aarav', 'Kumar', '2002-03-15', '9000000001', 'aarav@gmail.com'),
(6, 'Meera', 'Shah', '2005-07-22', '9000000002', 'meera@gmail.com'),
(7, 'Rohan', 'Patel', '2010-01-10', '9000000003', 'rohan@gmail.com'),
(8, 'Isha', 'Reddy', '2021-02-05', '9000000004', 'isha@gmail.com');

INSERT INTO Policies

VALUES

(1, 'Home Secure', 'Home Insurance', 25000, 10),
(2, 'Life Shield', 'Life Insurance', 18000, 20),
(3, 'Health Plus', 'Health Insurance', 12000, 5),
(4, 'Vehicle Safe', 'Vehicle Insurance', 15000, 3),
(5, 'Health Basic', 'Health Insurance', 11000, 1),
(6, 'Motor Secure', 'Motor Insurance', 10000, 1);

INSERT INTO Agents

VALUES

(1, 'Ramesh Kumar', '9123456789', 'Hyderabad'),
(2, 'Sita Rao', '9234567890', 'Bangalore'),
(3, 'Amit Singh', '9345678901', 'Delhi');

INSERT INTO PolicyAssignments

VALUES

(1, 1, 1, 1, '2022-01-01', '2032-01-01'),
(2, 2, 2, 2, '2021-06-15', '2041-06-15'),
(3, 3, 3, 3, '2023-03-10', '2028-03-10'),
(4, 4, 4, 1, '2024-02-01', '2027-02-01');

INSERT INTO Claims

VALUES

(1, 1, '2023-05-20', 50000, 'Approved'),
(2, 2, '2022-08-10', 100000, 'Pending'),

(3, 3, '2024-01-05', 25000, 'Rejected');

4.4.4 Select commands

--View all records Customers table.

SELECT * FROM Customers

--View all records of PolicyAssignment table with CustomerId, PolicyId, StartDate and EndDate columns only.

SELECT CustomerID,PolicyID,StartDate,EndDate FROM PolicyAssignments;

--Display all policies of Health type.

SELECT * FROM Policies WHERE PolicyType='Health Insurance';

--Display policies having premium amount more than 10000 and DurationYears is 1.

SELECT * FROM Policies WHERE PremiumAmount>10000 AND DurationYears=1;

--Display unique city names from where agents belong to.

SELECT DISTINCT City FROM Agents;

4.4.5 WHERE, IN, BETWEEN, LIKE

--using IN

SELECT * FROM Policies WHERE PolicyType IN ('Home Insurance','Motor Insurance','Life Insurance');

--using WHERE

SELECT * FROM Claims WHERE ClaimStatus='Rejected';

--using BETWEEN

SELECT * FROM Customers WHERE DateOfBirth BETWEEN '2001-01-01' AND '2022-12-31';

--using LIKE

SELECT * FROM Agents WHERE City LIKE '_a%';

4.4.6 AGGREGATE FUNCTIONS:

-- Max and Min claim amount

SELECT

MAX(ClaimAmount) AS MaxClaim,

MIN(ClaimAmount) AS MinClaim

```
FROM Claims;  
  
-- Latest claim  
  
SELECT TOP 1 *  
  
FROM Claims  
  
ORDER BY ClaimDate DESC;
```

4.4.7 UPDATE & DELETE:

```
-- Increase health policy premium by 10%  
  
UPDATE Policies  
  
SET PremiumAmount = PremiumAmount * 1.10  
  
WHERE PolicyType = 'Health';  
  
-- Delete expired policies  
  
DELETE FROM PolicyAssignments  
  
WHERE EndDate < GETDATE();
```

4.4.8 COMPUTED COLUMNS:

```
SELECT PolicyID,  
  
PolicyName,  
  
PremiumAmount,  
  
PremiumAmount * 0.06 AS LocalTax,  
  
PremiumAmount * 1.06 AS PremiumWithTax,  
  
(PremiumAmount * 1.06) / 12 AS MonthlyPremium  
  
FROM Policies;
```

4.4.9 Offset & Fetch:

```
SELECT * FROM Customers  
  
ORDER BY CustomerID  
  
OFFSET 1 ROWS  
  
FETCH NEXT 2 ROWS ONLY;
```

4.4.10 JOINS:

-- List of Policies Assigned to Customer

```
SELECT
    p.PolicyId,
    p.PolicyName,
    p.PolicyType,
    p.PremiumAmount
FROM PolicyAssignments pa
JOIN Policies p
    ON p.PolicyId = pa.PolicyId
```

WHERE pa.CustomerId = 5;

--Customers with Their Policies (Including Customers Without Policies)

```
SELECT
    c.CustomerId,
    c.FirstName,
    p.PolicyName
FROM Customers c
LEFT JOIN PolicyAssignments pa
    ON pa.CustomerId = c.CustomerId
LEFT JOIN Policies p
    ON p.PolicyId = pa.PolicyId;
```

4.4.11 GROUPING AND HAVING:

```
SELECT
    c.FirstName,
    SUM(cl.ClaimAmount) AS TotalClaimAmount
FROM Claims cl
JOIN PolicyAssignments pa
    ON pa.AssignmentId = cl.AssignmentId
JOIN Customers c
```

ON c.CustomerId = pa.CustomerId

GROUP BY c.FirstName

HAVING SUM(cl.ClaimAmount) > 50000;